

YASH BAVADIYA

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EDUCATION

Indian Institute of Technology, Jodhpur

Aug 2024 – Jul 2028

Bachelor's Degree - Computer Science and Engineering - CGPA - 7.21

Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL

AI/ML: LLM Prompting & Evaluation, Agentic Pipelines, Retrieval-Augmented Generation, FAISS Vector Search, Embeddings, Hugging Face Transformers, PyTorch, Model Quantization, Pandas, NumPy

Backend & Infrastructure: Node.js, Express, REST APIs, Third-Party API & Webhook Integration, Redis (pub/sub, BullMQ job queues), PostgreSQL, MongoDB, Prisma, Docker, Nginx, GitHub Actions, Linux, WebSockets

Tools & Frameworks: Git, React, Next.js, Qt (C++), Postman, Vercel, VS Code

Core Competencies: Data Structures and Algorithms, System Design, Distributed Systems, Operating Systems, Production Debugging, Technical Writing

EXPERIENCE

Google Summer of Code 2026

May 2026 – Aug 2026

Kdenlive @KDE Community | Open Source Contributor, C++ and Qt

Remote

- Selected for **GSoC 2026** among 1,200 global contributors to build **RGB curve, gradient, and speed ramp editors** for Kdenlive, a professional **video editing** suite, working in C++/Qt on creative and color-grading tooling.
- Shipped **20 contributions** across Kdenlive and upstream repositories, resolving an **11-year-old bug** via a gap-detection algorithm now running in production for **41,000 monthly Flathub installs**.

Undergraduate Researcher

Jul 2025 – Nov 2025

Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder

IIT Jodhpur

- Engineered an offline **Retrieval-Augmented Generation pipeline** in Python, integrating **FAISS vector search** with 4-bit quantized Phi-3 and Flan-T5-XL for document understanding and information extraction.
- Built an **LLM evaluation harness** benchmarking prompt and retrieval strategies across **8 ROUGE metrics** (+**71.77%** over baseline), hitting **3 second query latency** on 300-page PDFs across 3 feedback-driven iterations.

PROJECTS

DaemonDoc | [GitHub](#) | [Live](#) | TypeScript, LLM APIs, Express, MongoDB, Redis (BullMQ), Docker

Jan 2026 – May 2026

- Built an **LLM agent pipeline** on a **Redis (BullMQ)** queue, triggered by GitHub webhooks with idempotency and loop guards, sustaining an **85% success rate** across **959 production runs** over 68 repositories for **50 users**.
- Cut per-run token cost roughly **10x** with a two-stage router that sends file paths and token estimates, never file content, to a cheap model to select context before invoking the main model, with failover across 6 provider keys.
- Designed content-addressed section hashing and an output validator that blocks hallucinated or malformed sections, retrying in strict mode and skipping the commit rather than corrupting a user's README.

Flock | [GitHub](#) | [npm](#) | TypeScript, WebSocket, Redis, React, Docker

Feb 2026 – Jun 2026

- Released 3 npm packages enabling multiplayer cursors and live presence in web applications via a **10-line integration**, installable as a **55MB** self-hosted Docker image or npx.
- Implemented a horizontally scalable **Redis pub/sub relay** across server instances with keyspace-notification-driven eviction, reducing stale presence state to under **30 seconds** on ungraceful disconnects.
- Delivered cursor position smoothing with continuous mid-animation target updates, exponential backoff reconnection, and **58 tests** covering throttling, time-to-live eviction, and message delivery.

IIT Jodhpur PHC System | [GitHub](#) | Node.js, Express, PostgreSQL, Prisma, Docker

Jan 2026 – Apr 2026

- Spearheaded backend architecture replacing a paper workflow with a role-based platform of **63 REST endpoints** over a **20-model relational schema**, integrating third-party **LDAP** and JWT authentication across **6 roles**.
- Verified reliability via **48 test cases** and load tests at **20 concurrent users**, securing production deployment approval projecting **5 staff-hours saved per day**.

COURSEWORK

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|----------------------------------|-------------------------------|-------------------------------|
| • Machine Learning | • Probability and Statistics | • Software Engineering |
| • Pattern Recognition | • Database Management Systems | • Object-Oriented Programming |
| • Data Structures and Algorithms | • Operating Systems | • Discrete Mathematics |

ACHIEVEMENTS

- JEE Advanced 2024: AIR **3449**; JEE Main 2024: **99.65** percentile, AIR **5680**.
- Achieved Max Rating of **1429** on Codeforces; solved **450+** algorithmic problems on LeetCode in C++.
- Merged **84+ open-source contributions** across GitHub and GitLab, shipping reviewed, production features and bug fixes to **Kdenlive**, **Vitest**, **Wagtail**, and **croc**.

LEADERSHIP

- Junior Core Team Member** at DevUp Labs, IIT Jodhpur; mentored 30 students and facilitated 2 open-source workshops.
- Core Member** at E-Cell, IIT Jodhpur; supported 3 entrepreneurship events and initiatives over 9 months.
- Core Member** at Programming Club, IIT Jodhpur; co-organized 5 contests and 10 sessions on competitive programming.